Integrating Coupled Dictionary Learning and Distance Preserved Probability Distribution Adaptation for Multispectral–Hyperspectral Image Collaborative Classification

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***Abstract*— With the development of observation technology in remote sensing (RS), large-area multispectral (MS) images can be easily obtained. However, due to the limitation of imaging devices, only a limited range of hyperspectral (HS) images with higher spectral resolution can be obtained. This article mainly focuses on how to use limited HS images to improve the classification performance of MS images. In order to solve this problem, this article proposes an MS–HS image collaborative classification method, which integrates coupled dictionary learning and distance preserved probability distribution. First, image reconstruction based on coupled dictionary learning is performed, in which sparse representation and dictionary learning are used to generate HS images from MS images through spectral superresolution, so that the spectral features of the MS data and HS data are converted to the same feature space for feature space alignment. Second, the probability distribution is adapted, in which the marginal and conditional probabilities are adapted to further narrow the difference between the real HS data and the generated HS data. At the same time, the consistency of the data structure of the source domain before and after the mapping is maintained, so that the same class of data is more compact after the mapping and reduces the spacing within the same class. Compared with the state-of-the-art methods, this article conducts the experiments on three MS–HS RS datasets, which demonstrate the superiority of the proposed method.**

***Index Terms*— Classification, hyperspectral (HS) image, multispectral (MS) image, probability distribution adaptation, remote sensing (RS).**

I. Introduction

THE hyperspectral (HS) and multispectral (MS) remote sensing (RS) are important means to obtain information about ground features [1]. MS RS has been widely used in land analysis and utilization and environmental detection [2]. As a relatively new observation method, HS RS can observe the spectral information of the region of interest more deeply and becomes a powerful tool for detecting surface information, which is widely used in classification [3], [4], intrinsic image recovery [5], image segmentation [6], and object detection [7]. Compared with MS images, HS images have more bands and better spectral resolution, but their spatial resolution is lower than that of the MS images. In the classification problem, the quality of the obtained data directly determines the learning effect, but it is not a simple matter to collect high-quality data in practical applications. When performing land cover classification tasks, the images of large areas cannot be obtained due to the limitations of HS imaging equipment, but large-scale MS data (such as Sentinel-2) can be easily obtained for free [8]. However, since the spectral information of MS data is less than that of the HS data, the features cannot be expressed more effectively, resulting in poor classification results. To achieve large-area fine classification, it is necessary to consider the problem of collaborative classification in MS–HS images, which can be seen as a special kind of multimodal HS RS [9]–[12]. What is more, many studies focus on how to use the MS and HS images in the common coverage area for learning and training to improve the classification performance of large-area MS images.

At present, MS–HS images collaborative classification methods can be categorized into three categories, which are image fusion, image generation, and transfer learning. In order to improve the classification ability of MS images, some scholars fuse the corresponding MS–HS images [13]–[17]. For example, Wei *et al.* [18] presented a multiband image fusion algorithm with the linear spectral mixture model, which used the prior knowledge of spectral decomposition to fuse images to obtain more accurate reconstruction results. Similarly, Xu *et al.* [19] proposed the MS–HS images fusion method based on sparse representation of nonlocal patch tensors. However, due to the limitations of imaging equipment, HS data cannot be acquired in a wide range, which brings a huge challenge to the collection of MS–HS image pairs of the same size for the fusion task.

Some scholars reconstruct HS images with a limited range of MS–HS images to improve the classification ability of MS images. Recently, Sun *et al.* [20] investigated the weighted spectral angular distance (SAD) to filter the transformation

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matrix and used the transformation estimation to reconstruct the HS data. What is more, the joint sparse and low-rank learning (JSLoL) method was proposed, which used the common areas of HS images and MS images to learn low-rank overcomplete dictionaries and sparse representations [21]. Liu *et al.* [22] added sparse representation coefficient error and classification error constraints to the reconstruction dictionary. Moreover, the HS images that correspond to the nonoverlapped MS images are further reconstructed by using the shared sparsity coefficients obtained from the corresponding MS part. Specially, RGB images were used to restore HS images [23], [24]. However, estimating the simple regression matrix from the limited MS–HS pair or RGB images is a highly ill-conditioned problem, and there is still a certain difference in the distribution of the generated HS image spectrum and the real HS image spectrum.

There are also many methods based on transfer learning [25] to study this problem, aiming at transferring richer information from HS images to MS images, thereby improving the classification ability of MS images. Represented by the transfer component analysis (TCA) [26] method, many transfer learning algorithms have been proposed one after another [27]–[31]. The joint distribution adaptation (JDA) [27] further introduces conditional probability adaptation based on TCA. Zhang *et al.* [29] presented a guided transfer framework, which introduced a weighted matrix of the difference between the source subspace and the target subspace. A multiple geodesic flow kernel learning framework [30] is proposed to exploit both spatial and spectral features for unsupervised cross-temporal classification with bitemporal HS images. However, all the above methods solve the transfer learning problem with the same dimension of feature space [32], while the spatial-spectral dimensions of MS and HS images are different. Due to the different feature dimensions of MS–HS images, this problem can be regarded as a heterogeneous transfer learning problem. At present, there are many computer vision methods to study the heterogeneous transfer problem [33]. Tsai *et al.* [34] derived the domain invariant feature space to improve the matching between the source domain and the target domain. Long *et al.* [35] applied graph-based label propagation (GLP) technology for transfer learning and proposed the adaptation regularization based transfer learning (ARTL) method. In RS, the common subspace (Cospace) learning [36] method used two kinds of RS data, HS and MS, to jointly estimate a Cospace under the guidance of label information. Chen *et al.* [37] proposed a semisupervised transfer learning method based on nonnegative matrix factorization, which utilized the labeled samples of HS images in the source domain to assist in the classification of HS images obtained by different sensors. A mutation nuclear subspace learning algorithm [38] is exploited for domain adaptation, but it is prone to negative transfer problems if the source domain and target domain are too different. It is worth noting that the transfer effect of these methods mentioned above is not ideal when the source domain and target domain differ significantly. In addition, there are some methods based on deep transfer learning to research this problem, such as correlated neural networks [38] and autoencoder architectures. Although methods based on deep learning tend to obtain deep features, they require a large number of labeled samples.

In response to the above problems, this article proposes an MS–HS images collaborative classification method, which adapts distance preserved probability distribution after coupled dictionary learning. Specifically, only a small common part of MS-HS data is used to improve the classification ability of MS images in large scenes. Different from the existing methods, this method performs probability distribution adaptation while aligning the MS–HS data space by reconstructing HS images and combines structural maintenance constraints to achieve the best results. The proposed method makes full use of the characteristics of MS and HS images and reduces the difference between the generated HS image and the real HS image, thereby improving the classification ability of MS images. In summary, the main contributions of this article are summarized as follows.

1) An MS–HS image collaborative classification method is proposed, which transfers limited information in the public area of MS–HS images to classify MS images and has great potential in the high precision classification of large scenes.
2) This article combines image reconstruction and probability distribution adaptation, which makes full use of the advantages of image generation methods and transfer learning methods, and proposes a distribution adaptation algorithm based on distance preservation. While adapting the source and target domain data, distance preservation constraints are added to make the adaptation effect better, thereby improving the classification capabilities of MS images.

The rest of this article is organized as follows. In Section II, the architecture of the proposed algorithm model is introduced. Section III gives the experimental results and comparisons with other methods. Finally, the proposed method is summarized in Section IV.

II. Proposed Algorithm

In this section, our motivation and idea are introduced briefly. Accordingly, the basic implementation process of the proposed method and optimization is described in detail. Fig. 1 illustrates the holistic workflow of our method.

A. Motivation

In the RS image classification problem, HS data have more spectral information than MS data and can express features effectively. However, due to the limitation of HS imaging equipment, large-area images cannot be obtained. With the development of observational technology, large-scale MS data can be easily obtained for free. If the advantages of MS and HS data can be combined, it will be helpful for the fine classification of large-scale RS scenes. So, the main goal of this article is to effectively utilize the common coverage area of MS–HS images. Transferring the HS information to the MS image makes the spectral characteristics of the reconstructed large-scale HS data more representative, so as to achieve better

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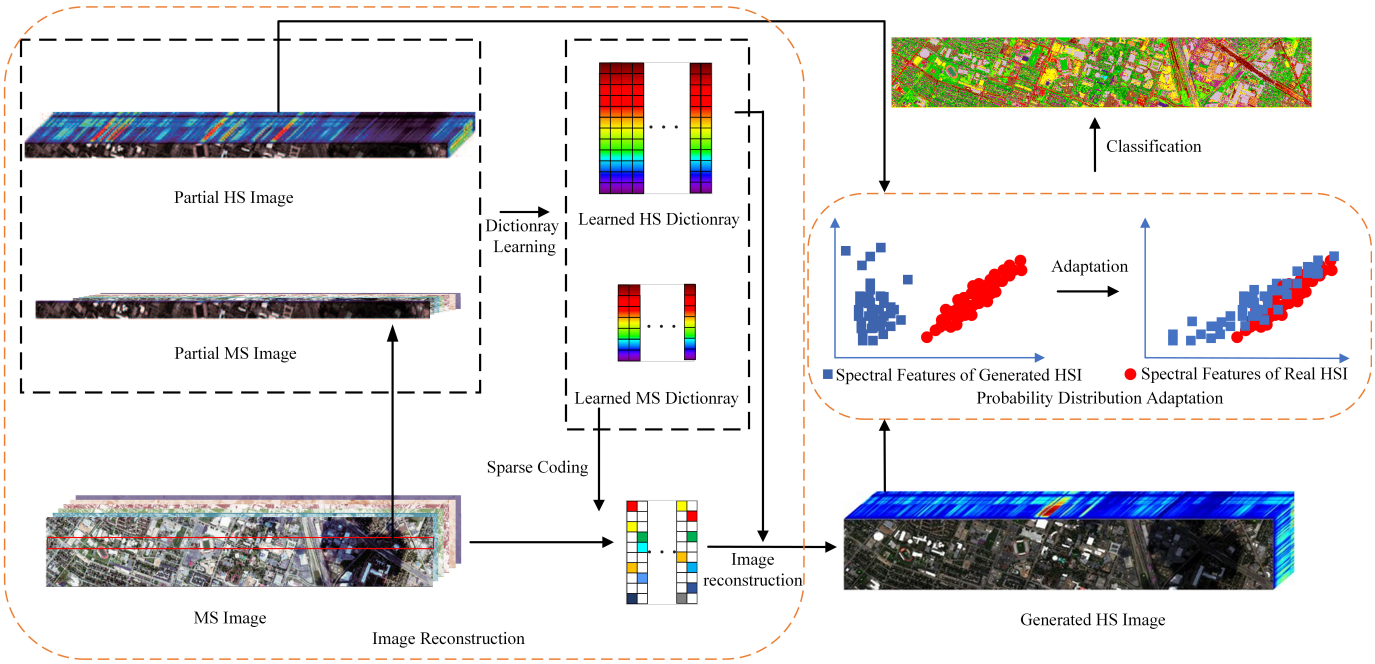


Fig. 1. Proposed model for the classification task. It consists of two parts that are image reconstruction and probability distribution adaptation.

classification results. For this purpose, we propose an MS–HS image collaborative classification method, which integrates the coupled dictionary learning and distance preserved probability distribution adaptation.

B. Method Overview

The basic framework of the method proposed in this article is shown in Fig. 1. It can be seen that there are two core parts, which are the HS reconstruction module and probability distribution adaptation module. First, spectral superresolution on the MS image is performed to convert the MS and HS images into the same feature space. Then, probability adaptation is performed on the generated HS image and the real HS image to map their distribution to the same space. Finally, the classification result of the large-scale MS images is obtained through the classifier.

In the image reconstruction stage, first, low-rank dictionary learning is performed through the MS–HS data covering the same area, and the MS dictionary $\mathbf{B}_{\mathrm{m}}$ and HS dictionary $\mathbf{B}_h$ are learned from the common small area. Then, the MS dictionary $\mathbf{B}_{\mathrm{m}}$ is used to learn the sparse coefficients $\mathbf{Q}$ for the global MS data and generate HS data through the sparse coefficient of the MS data and the HS dictionary. This completes the alignment of the features of the MS data and the HS data.

The distribution adaptation stage mainly expresses the generated spectral features and the real spectral features uniformly in distribution. Since there is still a certain difference between the HS data generated by dictionary learning and the real HS data, it is necessary to adapt the probability of the two spectral features. The marginal distribution and the conditional distribution of the spectral features of the real HS data and that of the generated HS data are adapted, and the distance constraint is added to the mapping. In order to adapt the spectral feature distribution of the two, the MS image can obtain a more distinctive feature representation and achieve a better classification effect. We will describe these steps in the method in detail below.

C. Image Reconstruction Module

In the first step, the dictionary learning and sparse representation are used to reconstruct the MS image into an HS image. Since part of the MS and HS images used in this article has the same coverage area, HS and MS images can be expressed by multiplying the same set of sparse coefficients with HS and MS dictionaries. Therefore, this idea is used to realize the spectral superresolution reconstruction of MS images. In the overlapped part of HS and MS images, let $\mathbf{D}_M = [\mathbf{x}_m^1, \ldots, \mathbf{x}_m^i, \ldots, \mathbf{x}_m^{N_1}] \in \mathbf{R}^{d_M \times N_1}$ be an MS matrix with d_M bands by N_1 pixels, $\mathbf{D}_S = [\mathbf{x}_s^1, \ldots, \mathbf{x}_s^i, \ldots, \mathbf{x}_s^{N_1}] \in \mathbf{R}^{d_H \times N_1}$ be an HS matrix with d_H bands by N_1 pixels, and $\mathbf{D}_T = [\mathbf{x}_t^1, \ldots, \mathbf{x}_t^i, \ldots, \mathbf{x}_t^{N_1}] \in \mathbf{R}^{d_H \times N_1}$ be the generated HS matrix. $\mathbf{D}_{M_g} = [\mathbf{x}_{m_g}^1, \ldots, \mathbf{x}_{m_g}^i, \ldots, \mathbf{x}_{m_g}^{N_2}] \in \mathbf{R}^{d_M \times N_2}$ denotes the MS matrix and $\mathbf{D}_{T_g} = [\mathbf{x}_{t_g}^1, \ldots, \mathbf{x}_{t_g}^i, \ldots, \mathbf{x}_{t_g}^{N_2}] \in \mathbf{R}^{d_H \times N_2}$ denotes the generated HS matrix in the global region with d_M and d_H bands by N_2 pixels. HS images and MS images can be sparsely represented by the following formulas:

$$\mathbf{D}_S = \mathbf{B}_h \mathbf{P} \tag{1}$$

$$\mathbf{D}_M = \mathbf{B}_m \mathbf{P} \tag{2}$$

where $\mathbf{B}_m$ and $\mathbf{B}_h$ are the low-rank dictionaries which will be learned from MS and HS data. $\mathbf{P}$ is a uniform sparse representation on the two dictionaries. It can be solved by the following objective function:

$$\min_{\mathbf{B}_h, \mathbf{B}_m, \mathbf{P}} \|\mathbf{D}_S - \mathbf{B}_h \mathbf{P}\|_{\mathrm{F}}^2 + \|\mathbf{D}_M - \mathbf{B}_m \mathbf{P}\|_{\mathrm{F}}^2 + \|\mathbf{P}\|_{1,1} \tag{3}$$

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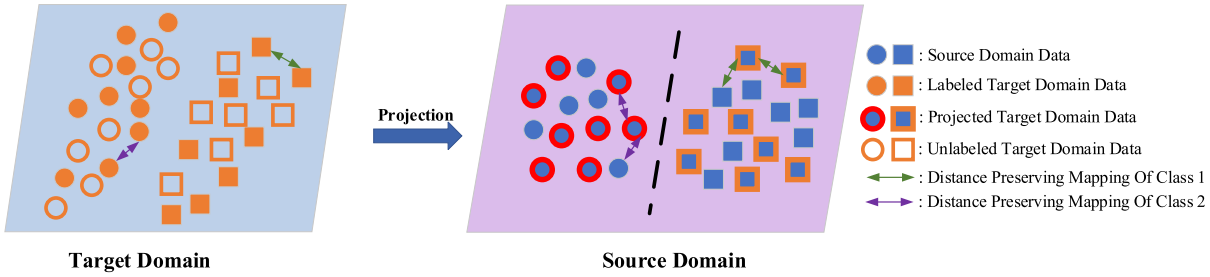


Fig. 2. Illustration of probability distribution adaptation for the proposed method.

where $\|\cdot\|_{\mathrm{F}}$ and $\|\cdot\|_{1,1}$ represent F norm and 1 norm, respectively. By learning $\mathbf{B}_m$, $\mathbf{B}_h$, and $\mathbf{P}$, we can use the MS dictionary to learn the unified coefficient matrix sparse coefficients $\mathbf{Q}$ of the global MS–HS image

$$\mathbf{D}_{M_g} = \mathbf{B}_m \mathbf{Q}. \tag{4}$$

It can be expressed as follows:

$$\min_{\mathbf{Q}} \left\|\mathbf{D}_{M_g} - \mathbf{B}_m \mathbf{Q}\right\|_{\mathrm{F}}^2 + \|\mathbf{Q}\|_{1,1}. \tag{5}$$

Then, the sparse coding and HS dictionary can be used to reconstruct a large area of the HS image

$$\mathbf{D}_{T_g} = \mathbf{B}_h \mathbf{Q}. \tag{6}$$

Based on the above theory, Gao *et al.* [21] add constraints to the dictionary and sparseness and propose the JSLoL algorithm. What is more, in order to obtain the discrimination $\mathbf{P}$ and make the coupling dictionary useful for classification, Liu *et al.* [22] develop a class-guided coupled dictionary learning (CGCDL) method that introduced two discriminant sparse coefficient errors and classification errors in the objective function. In this way, we can get $\mathbf{D}_{T_g}$ and $\mathbf{D}_T$ from $\mathbf{D}_{M_g}$ and $\mathbf{D}_M$, respectively.

D. Probability Distribution Adaptation Module

In order to further reduce the difference between the generated HS $\mathbf{D}_{T_g}$ and real HS data $\mathbf{D}_S$, they are probabilistically adapted. The real HS data $\mathbf{D}_S$ are used as the source domain data, and the generated HS data $\mathbf{D}_{T_g}$ are used as the target domain data. What is more, $\mathbf{D}_T$ and $\mathbf{D}_S$ are in the overlapped part, and the labels are $\mathbf{Y}_T$ and $\mathbf{Y}_S$ of the target domain and source domain, respectively.

JDA [27] has been successfully applied to domain adaptation tasks. Inspired by recent domain adaptation methods, this article learns a feature transformation $\mathbf{T}$, which projects target domain data into a subspace derived from source domain data. In this derived subspace, the goal of this article is to correlate the generated HS data of the same feature dimension with the different distributions of the real HS data. This association is achieved by matching the data distribution [i.e., the marginal distribution $\mathrm{P}_T(\mathbf{T}^{\mathrm{T}}\mathbf{D}_T)$ and $\mathrm{P}_S(\mathbf{D}_S)$ and the conditional distribution $\mathrm{P}_T(\mathbf{Y}_T|\mathbf{T}^{\mathrm{T}}\mathbf{D}_T)$ and $\mathrm{P}_S(\mathbf{Y}_S|\mathbf{D}_S)$] to eliminate the difference between the two domains. Since only reducing the difference in marginal distributions does not guarantee that the conditional distributions between domains can also be narrowed, it is necessary to adapt the marginal distributions and conditional distributions at the same time. In particular, this article proposes structure preservation constraints on the target domain data to improve the classification ability after the two domain probability adaptations are completed.

The concept of probability distribution adaptation is illustrated in Fig. 2. Among them, the blue and pink scenes represent the target domain and the source domain, and the circles and squares represent two categories, respectively. The conditional probabilities and marginal probabilities of the source and target domains are adapted through probability adaptation, and the inner distance of classes can be constrained while adapting. Specifically, the distance between the mapped target domain and the target domain and the distance between the mapped target domain and the source domain of the same category are reduced through constraints, so as to achieve a better adaptation effect and make the classification result more accurate. Specifically, probability adaptation and local mapping maintenance are used to reduce the difference between the two domains, thereby solving problems such as spectral drift. The problem can be formulated as follows:

$$\min_{\mathbf{T}} \ \mathbf{E}_M(\mathbf{T}, \mathbf{D}_S, \mathbf{D}_T) + \mathbf{E}_C(\mathbf{T}, \mathbf{D}_S, \mathbf{D}_T) + \mathbf{E}_T(\mathbf{T}, \mathbf{D}_T) + \lambda\|\mathbf{T}\|^2 \tag{7}$$

where the first and second terms ($\mathbf{E}_M$ and $\mathbf{E}_C$), respectively, measure the difference between the marginal distribution and conditional distribution of the real HS and generated HS data. $\mathbf{E}_T$ denotes the term of label and structure-preserving in generated HS images. The last item is to prevent overfitting $\mathbf{T}$.

In order to reduce the difference between the marginal distributions $\mathrm{P}_T(\mathbf{T}^{\mathrm{T}}\mathbf{D}_T)$ and $\mathrm{P}_S(\mathbf{D}_S)$, this article follows [26], [27] and uses the empirical maximum mean difference (MMD) as a distance metric to compare different distributions. Therefore, the marginal distribution can be expressed as follows:

$$\mathbf{E}_M(\mathbf{T}, \mathbf{D}_S, \mathbf{D}_T) = \left\| \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbf{T}' \mathbf{x}_t^i - \frac{1}{n_s} \sum_{i=1}^{n_s} \mathbf{x}_s^i \right\|^2 \tag{8}$$

where $\mathbf{x}_t^i$ indicates the labeled generated HS data, and n_t denotes the number of labeled generated HS data. Similarly, $\mathbf{x}_{\mathrm{s}}^i$ is the real HS data, and n_{s} represents the number of it.

In addition to adapting the marginal distribution, minimizing the difference between the conditional distributions $\mathrm{P}_T(\mathbf{Y}_T|\mathbf{T}^{\mathrm{T}}\mathbf{D}_T)$ and $\mathrm{P}_S(\mathbf{Y}_S|\mathbf{D}_S)$ is essential for the robustness

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of the adaptation of the real HS data and the generated HS data. However, matching conditional distributions is difficult, even if the distribution statistics between the two domains are fully explored. Therefore, a major computational problem is to reduce the difference of the distribution by explicitly minimizing the appropriate distance metric. Since the posterior probabilities $P_T(\mathbf{Y}_T|\mathbf{T}^{\mathrm{T}}\mathbf{D}_T)$ and $P_S(\mathbf{Y}_S|\mathbf{D}_S)$ are very complicated, it is impossible to estimate them directly. This article refers to the solution in [30] and uses Bayes' Theorem to estimate $P_T(\mathbf{T}^{\mathrm{T}}\mathbf{D}_T|\mathbf{Y}_T)$ and $P_S(\mathbf{D}_S|\mathbf{Y}_S)$. Therefore, the conditional distribution can be expressed as follows:

$$\mathbf{E}_C(\mathbf{T},\mathbf{D}_S,\mathbf{D}_T)=\sum_{c=1}^{C}\left\|\frac{1}{n_t^{\mathrm{c}}}\sum_{i=1}^{n_t^{\mathrm{c}}}\mathbf{T}'\mathbf{x}_t^{i,c}-\frac{1}{n_{\mathrm{s}}^{\mathrm{c}}}\sum_{i=1}^{n_{\mathrm{s}}^{\mathrm{c}}}\mathbf{x}_s^{i,c}\right\|^2+\frac{1}{n_{\mathrm{s}}^{\mathrm{c}}n_t^{\mathrm{c}}}\sum_{i=1}^{n_t^{\mathrm{c}}}\sum_{i=1}^{n_{\mathrm{s}}^{\mathrm{c}}}\left\|\mathbf{T}'\mathbf{x}_t^{j,c}-\mathbf{x}_s^{i,c}\right\|^2. \quad (9)$$

In order to make the generated HS data of the same category closer after being mapped to the source domain, this article proposes a constraint on the structure of the target domain data, and the expression is shown in the following:

$$\mathbf{E}_T(\mathbf{T},\mathbf{D}_T)=\sum_{i=1}^{n_t^{\mathrm{c}}}\sum_{j=1}^{n_t^{\mathrm{c}}}\omega_{ij}\left\|\mathbf{T}'\mathbf{x}_t^{i}-\mathbf{T}'\mathbf{x}_t^{j}\right\|^2 \quad (10)$$

where $\omega_{ij}=\begin{cases}e^{-\left\|x_t^i-x_t^j\right\|^2/\sigma} & \left\{x_t^i,x_t^j\in x_t^c\right\}\\ 0\end{cases}$ is the weight for distance. When minimizing (10), if two generated HS data in the target domain belong to the same category, then the penalty for them will be increased, forcing them to be closed after being mapped to the source domain and vice versa.

In conclusion, the proposed method combining probability adaptation and distance preservation is calculated as follows:

$$\begin{aligned}\min_{\mathbf{T}}&\left\|\frac{1}{n_t}\sum_{i=1}^{n_t}\mathbf{T}'\mathbf{x}_t^{i}-\frac{1}{n_s}\sum_{i=1}^{n_s}\mathbf{x}_s^{i}\right\|^2\\&+\sum_{c=1}^{C}\left\|\frac{1}{n_t^{\mathrm{c}}}\sum_{i=1}^{n_t^{\mathrm{c}}}\mathbf{T}'\mathbf{x}_t^{i,c}-\frac{1}{n_{\mathrm{s}}^{\mathrm{c}}}\sum_{i=1}^{n_{\mathrm{s}}^{\mathrm{c}}}\mathbf{x}_s^{i,c}\right\|^2\\&+\frac{1}{n_{\mathrm{s}}^{\mathrm{c}}n_t^{\mathrm{c}}}\sum_{i=1}^{n_t^{\mathrm{c}}}\sum_{i=1}^{n_{\mathrm{s}}^{\mathrm{c}}}\left\|\mathbf{T}'\mathbf{x}_t^{j,c}-\mathbf{x}_s^{i,c}\right\|^2+\sum_{i=1}^{n_t^{\mathrm{c}}}\sum_{i=1}^{n_t^{\mathrm{c}}}\omega_{ij}\left\|\mathbf{T}'\mathbf{x}_t^{i}-\mathbf{T}'\mathbf{x}_t^{j}\right\|^2\\&+\lambda\|\mathbf{T}\|^2. \quad (11)\end{aligned}$$

E. Model Optimization

In the image reconstruction module, the objective functions of JSLoL [21] and CGCDL [22] can be optimized and solved by alternating direction method of multipliers (ADMM).

In the probability distribution adaptation module, to solve the minimization of (11), taking the derivative of (11) with respect to $\mathbf{T}$ and setting the derivative to 0, the expression of $\mathbf{T}$ can be derived as follows:

$$\mathbf{T}=(\lambda\mathbf{I}_{dT}+\mathbf{X}_T(\mathbf{H}_{TM}+\mathbf{H}_{TC}+\mathbf{S})\mathbf{X}_T')^{-1}\times\left(\mathbf{X}_T(\mathbf{H}_{SM}+\mathbf{H}_{SC})\mathbf{X}_S'\right) \quad (12)$$

where

$$(\mathbf{H}_{TM})_{ij}=\frac{1}{n_tn_t},(\mathbf{H}_{SM})_{ij}=\frac{1}{n_sn_t},(\mathbf{W})_{ij}=\omega_{ij},(\mathbf{H}_{TC})_{ij}=\begin{cases}(1+n_t^c)/n_t^cn_t^c, & i,j\in C_i,i=j\\ 1/n_t^cn_t^c, & i,j\in C_i,i\neq j\\ 0, & \text{else}\end{cases}$$

$$(\mathbf{H}_{SC})_{ij}=\begin{cases}2/n_s^cn_t^c, & i,j\in C_i\\ 0, & \text{else}\end{cases}$$

$$\mathbf{S}=\mathbf{D}-\mathbf{W},\quad(\mathbf{D})_{ii}=\sum_j\omega_{ij}.$$

Through the mapping matrix $\mathbf{T}$, the generated HS data can be probabilistically adapted to the real HS data, and the difference between the two can be further reduced. Finally, the real HS data can be used for training to classify the mapped generated HS data.

III. Experimental Results and Discussion

In this section, we report the results of the experiments to evaluate the effectiveness of the proposed algorithm.

A. Data Description

In the experimental part, three sets of MS–HS datasets are utilized to verify the proposed algorithm, which are the University of Houston (UH) dataset and two Yellow River Estuary datasets. There is the detailed introduction to the three datasets below.

1) University of Houston: The UH dataset scene with 349 × 1905 pixels and 2.5-m spatial resolution comes from the UH in the United States and was used in the 2013 IEEE GRSS data fusion competition. The HS image with 144 bands was collected by the ITRES-Compact Airborne Spectrographic Imager (CASI) sensor between 17:37:10 and 17:39:50 on June 23, 2012, and the spectrum ranges from 380 to 1050 nm, including visible light and near-infrared bands. By using the MS spectral response function (SRF) of Sentinel-2 as a filter, the HS image is downsampled in the spectral domain to generate an MS image with a size of 349 × 1905 × 10 [36]. The MS image and part of HS image RGB false color composite image and training and test labels are shown in Fig. 3, and Table I lists the number of training and test samples. The objects in the observation scene are divided into 15 categories, which are healthy grass, sparse grass, artificial grass, trees, land, water, residential buildings, commercial buildings, roads, highways, railways, parking lots 1, parking lots 2, tennis courts, and running tracks.

2) Yellow River Estuary_1: The Yellow River Estuary dataset is composed of the GF5 Yellow River Estuary HS data with 30-m spatial resolution and the simulated GF1 data generated by GF5, which downsampled the spectrum according to the SRF of GF1. The spatial size is 734 × 763 of the GF5 image. The spectral range is 0.390–2.513 μm of the GF5 HS data, which contained 330 bands and was captured on November 1, 2018. The GF1 MS data generated by the GF5 data simulation have a total of four bands and the spectral range is 0.45–0.89 μm. The spatial resolution of simulated

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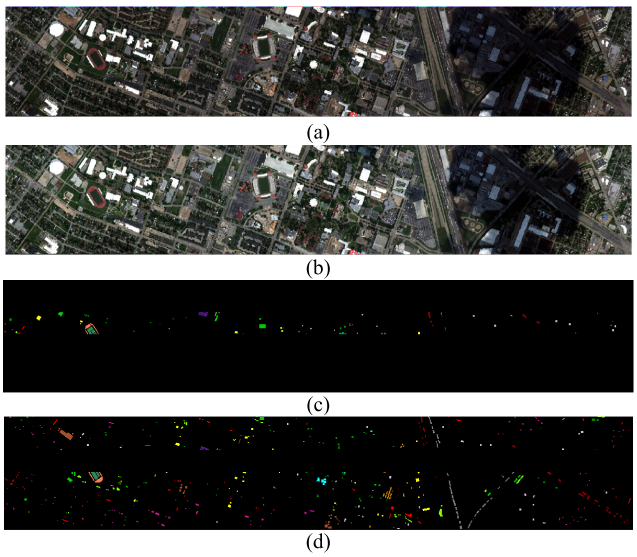


Fig. 3. UH dataset. (a) HS false color composite image. (b) MS false color composite image. (c) Training labels. (d) Test labels.

TABLE I
INFORMATION FOR THE UH DATASET

Class No.	Color	Class Name	Training	Test
1		Healthy Grass	537	699
2		Stressed Grass	61	1154
3		Synthetic Grass	340	357
4		Tree	209	1035
5		Soil	74	1168
6		Water	22	303
7		Residential	52	1203
8		Commercial	320	924
9		Road	76	1149
10		Highway	279	948
11		Railway	33	1185
12		Parking Lot1	329	904
13		Parking Lot2	20	449
14		Tennis Court	266	162
15		Running Track	279	381
Total			2897	12021

TABLE II
INFORMATION FOR THE YELLOW RIVER ESTUARY_1 DATASET

Class No.	Color	Class Name	Training	Test
1		Reed	252	554
2		Tamarix	398	143
3		Tidal Flat Reed	124	105
4		Salt Marshe	203	689
5		Suaeda Salsa	120	167
6		Mudflat	142	707
7		Water	237	1752
8		Spartina Alterniflora	239	965
Total			1715	5082

GF1 is 30 m/pixel and the spatial size is 734 × 763. The spatial size of the selected training data is 278 × 309. The objects in the observation scene are divided into eight categories, which are reed, tamarix, tidal flat reed, salt marshe, suaeda salsa, mudflat, water, and spartina alterniflora. Fig. 4 shows the images and training and test labels for GF1 and GF5. The specific settings of the label are shown in Table II.

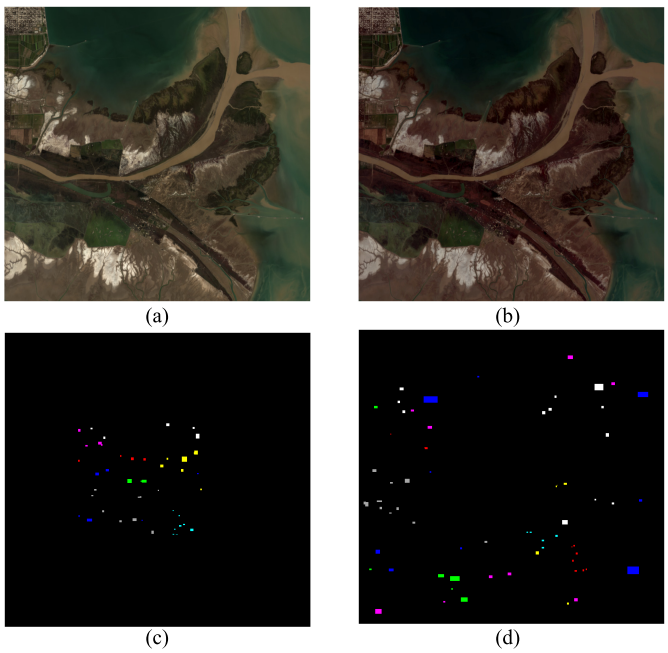


Fig. 4. Yellow River Estuary_1. (a) HS false color composite image. (b) MS false color composite image. (c) Training labels. (d) Test labels.

TABLE III
INFORMATION FOR THE YELLOW RIVER ESTUARY_2 DATASET

Class No.	Color	Class Name	Training	Test
1		Reed	559	1185
2		Tamarix	845	291
3		Tidal Flat Reed	269	211
4		Salt Marshe	430	1475
5		Suaeda Salsa	265	360
6		Mudflat	283	1492
7		Water	511	3736
8		Spartina Alterniflora	502	2031
Total			3664	10781

3) Yellow River Estuary_2: This dataset contains an MS image captured by the wide field view (WFV) sensor of GF1 Satellite on October 1, 2018, and the same HS image as the Yellow River Estuary_1 dataset. The GF1 data with 1066 × 1108 pixels have four bands and the spatial resolution of it is 16 m/pixel. The GF5 data have been upsampled to the same spatial resolution of GF-1 data by nearest neighbor (NN) interpolation. The false color images and training and test labels look the same as the Yellow River Estuary_1 dataset. There are also eight categories in the Yellow River Estuary_2 dataset. The spatial size of the selected training data is 404 × 448, and the specific settings of the label are shown in Table III.

B. Experimental Setting

This article uses overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ) to evaluate the results. In order to effectively verify our proposed method, several state-of-the-art methods related to our task are selected for comparison. The NN classifier is adopted to quantify the experimental results. Among them, the training and test data of baseline methods baseline-MSI (B-MSI) and baseline-HSI

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(B-HSI) are in the same domain. Specifically, B-HSI is trained and tested on the HS data, while B-MSI is trained and tested on the MS image. JSLoL and CGCDL use the real HS data for training and HS data generated by MS data for testing. The penalty parameters are set according to [21] and [22]. Cross-domain landmark selection (CDLS) aligns MS data and real HS data through transfer learning to classify. At the beginning, we set $\delta = 0.5$ and $\lambda = 0.5$ in the CDLS experiment. JSLoL-CDLS (J-CDLS) and CGCDL-CDLS (C-CDLS), respectively, use JSLoL and CGCDL methods to generate HS images from MS images and then adapt to the marginal probabilities of both, and Cospace learns an MS–HS common space for classification after mapping. Initially, we set the related optimal parameter $(\alpha, \beta, d) = (0.01, 0.01, 30)$ in this experiment. Furthermore, the proposed method (J-ours and C-ours) first uses dictionary learning to generate HS images, aligns the spectral spaces of the MS and HS images, and then probabilistically adapts the generated HS images and the real HS images to reduce the difference between them. In the HS reconstruction module, the setting of parameters follows JSLoL and CGCDL, and we set $\lambda = 0.6$ in the probability distribution adaptation module.

The parameter setting of different algorithms plays an important role in the performance analysis. In order to ensure the fairness of the experiment, the parameters of the comparison algorithm in the experiment are initially set according to the paper of this method. For the image reconstruction methods, the relevant parameters of image reconstruction have been analyzed in JSLoL and CGCDL methods, and parameter settings that are stable for most datasets have been selected. This article directly adopts their parameter settings. For other comparison methods, the parameter settings in these papers are first adopted and then fine-tuned for different datasets. At last, the best result of these methods in the experiment is taken as the final experiment result. Later, the parameter sensitivity will be analyzed in Section III-D for verifying the choice of parameters.

C. Experimental Results and Analysis

This article uses the NN classifier to experiment on three datasets, which are Houston HS data and simulated Sentinel-2 MS image, GF5 HS image and simulated GF1 MS image, GF5 HS image, and GF1 MS image. Tables IV–VI list the detailed results of the compared algorithms on three different MS-HS datasets. In general, the classification performance of the baseline method B-MSI is worse than that of B-HSI, and OA, AA, and Kappa are much lower because MS images cannot distinguish the types of features well owing to the lack of abundant spectral information. The baseline method B-HSI has the best classification performance on the two Yellow River Estuary datasets, and OA, AA, and Kappa are the highest because the HS image contains rich spectral information, which makes it easier to distinguish features. Compared to the Yellow River Estuary_1 dataset, Yellow River Estuary_2 has better classification performance. It may be due to the fact that the real MS data contain more information than the simulated data and achieve a better transfer effect. The performance of the proposed J-ours and C-ours methods is better than the other methods, which indicates that they can learn more discriminative feature representations and find better decision boundaries. More intuitively, the method proposed in this article carries out image generation and probability adaptation at the same time and combines structural preservation constraints to achieve the best classification accuracy. Compared with the J-ours method, C-ours adds the classification error to constrain when generating the HS image, so it achieves the best effect. As a method of image generation, JSLoL only uses the low-rank overcomplete dictionary and sparse representation to reconstruct HS images and it aligns the two feature spaces. In the dictionary learning process of the CGCDL method, two regularization terms, discriminant sparse coefficient error and classification error, are added to the objective function, so that the learned coupling dictionary is both representative and discriminative, so it is better than the JSLoL method. However, there is still a certain difference between the generated HS image and the real HS image through the common part of the spectral information. What is more, it can be seen from Table IV that the performance of CDLS based on transfer learning is also better than the baseline method and image generation methods. Notably, CDLS uses transfer learning to reduce the difference between the source domain and the target domain through probability adaptation, which can indicate that a limited amount of HS data can guide the corresponding MS data to generate better discriminative features. Accordingly, J-CDLS and C-CDLS use JSLoL and CGCDL, respectively, to generate HS images and then adapt the marginal probabilities of the two to further reduce the differences between the source domain and the target domain. Unlike them, Cospace learns the latent subspace by aligning different modalities and realizes the mapping between the source domain and the target domain, but Cospace does not consider probability adaptation. Therefore, combined with theoretical analysis, the experimental results are more reasonable. Therefore, in this problem, the proposed method can achieve a better classification effect. Experiments verify the effectiveness of the proposed method.

As shown in Fig. 5, in order to visually highlight the classification differences of different methods, the Houston data classification result map is displayed. Compared with other methods, C-ours can better transfer HS information to MS data through spectral feature alignment and adaptation. So as to obtain a more recognizable feature expression, the methods after probability adaptation have better classification results. The classification results of the proposed method are more reasonable. For example, the separation of farmland and row spacing crops is more accurate. Compared with other methods such as Cospace, C-ours has more advantages in the classification of parking lot 1 and parking lot 2. As shown in Table I, this article only selects limited samples in the center of the data for training in the Houston dataset. Unlike the two Yellow River Estuary datasets, the selection of training data in the Houston dataset causes the problem of imbalance of training samples. From Table I, it can be seen that the number of Healthy Grass and Synthetic Grass is dozens of times the number of categories, such as Residential, Road, and Railway. When the number of samples differs greatly, the classifier will

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TABLE IV
CLASSIFICATION RESULTS ON THE UH DATASET

Methods	B-HSI	B-MSI	JSLoL	CGCDL	CDLS	Cospace	J-CDLS	C-CDLS	J-ours	C-ours
Healthy Grass	68.81	67.67	67.81	64.66	74.24	84.41	92.42	73.53	**93.71**	92.84
Stressed Grass	77.64	77.82	77.56	71.32	79.98	69.76	89.86	77.04	**90.55**	89.70
Synthetic Grass	**100.0**	**100.0**	**100.0**	**100.00**	**100.0**	**100.0**	**100.00**	**100.0**	**100.00**	**100.0**
Tree	89.86	90.24	86.86	72.37	97.97	93.24	88.50	**99.32**	93.72	92.37
Soil	99.32	99.14	99.32	97.86	98.97	99.06	99.40	99.40	99.57	**99.49**
Water	86.14	86.14	86.14	**87.98**	83.82	84.49	81.85	82.84	86.47	87.47
Residential	45.05	41.23	39.73	57.27	60.43	68.83	68.91	**80.55**	68.91	74.98
Commercial	56.17	51.73	38.42	**61.80**	59.85	57.47	56.60	58.22	59.20	59.47
Road	65.54	65.01	65.54	62.05	63.27	65.45	59.18	**68.84**	57.53	58.86
Highway	47.36	46.31	42.72	42.72	54.21	45.57	50.95	60.64	53.69	**60.65**
Railway	36.29	39.49	37.81	33.25	40.50	**46.50**	38.73	37.13	44.64	44.56
Parking Lot1	43.47	41.70	40.38	45.58	40.04	48.34	55.09	42.48	63.83	**64.39**
Parking Lot2	0.45	0.45	0.45	0.67	0.44	0.67	0.67	0.00	0.89	**1.78**
Tennis Court	99.38	97.53	97.53	97.53	99.38	98.77	99.38	98.77	98.15	**99.38**
Running Track	97.38	97.64	97.11	**98.43**	97.37	97.38	97.64	97.90	97.64	97.48
OA	64.09	63.37	61.39	62.61	64.01	68.17	69.65	70.37	71.86	**72.42**
AA	67.52	66.80	65.15	66.23	70.03	70.667	71.94	71.77	73.91	**74.89**
Kappa	0.6101	0.6022	0.5808	0.5940	0.6121	0.6542	0.6703	0.6777	0.6944	**0.6984**

TABLE V
CLASSIFICATION RESULTS ON THE YELLOW RIVER ESTUARY_1 DATASET

Methods	B-HSI	B-MSI	JSLoL	CGCDL	CDLS	Cospace	J-CDLS	C-CDLS	J-ours	C-ours
Reed	**96.39**	39.53	47.47	38.45	40.25	38.45	35.74	34.84	41.16	40.43
Tamarix	**86.01**	78.32	79.02	82.22	72.03	82.52	80.42	83.01	83.22	83.22
Tidal Flat Reed	97.14	98.10	96.19	**100.00**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
Salt Marshe	99.85	100.0	100.0	**100.00**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
Suaeda Salsa	88.02	97.60	97.01	**100.00**	**100.0**	**100.0**	98.80	**100.0**	**100.0**	99.40
Mudflat	83.88	64.92	59.41	87.98	99.58	81.75	99.58	99.72	97.89	**99.79**
Water	**100.0**	99.49	**100.0**	**100.00**	**100.0**	99.60	**100.0**	**100.0**	**100.0**	**100.0**
Spartina Alterniflora	**85.49**	55.85	47.98	35.34	28.08	42.38	37.72	40.52	41.66	53.78
OA	93.74	79.24	77.98	78.86	78.98	79.18	80.51	81.09	82.11	**84.14**
AA	92.10	79.22	78.38	80.49	79.99	80.58	81.53	82.26	82.99	**84.58**
Kappa	0.9215	0.7440	0.7288	0.7391	0.7402	0.7431	0.7589	0.7659	0.7736	**0.8031**

TABLE VI
CLASSIFICATION RESULTS ON THE YELLOW RIVER ESTUARY_2 DATASET

Methods	B-HSI	B-MSI	JSLoL	CGCDL	CDLS	Cospace	J-CDLS	C-CDLS	J-ours	C-ours
Reed	**96.20**	45.49	47.09	54.35	55.95	55.70	51.48	57.55	58.90	64.30
Tamarix	**86.25**	72.85	74.91	52.23	57.04	64.60	67.35	75.26	74.23	85.22
Tidal Flat Reed	96.67	97.63	96.68	95.26	96.68	97.63	97.63	97.16	**97.63**	96.68
Salt Marshe	99.93	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
Suaeda Salsa	**88.33**	34.72	53.06	47.22	63.06	56.11	44.17	44.44	48.89	52.50
Mudflat	83.45	92.02	57.64	83.04	90.48	94.30	96.25	93.16	**99.46**	98.86
Water	**100.00**	99.57	**100.00**	97.35	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
Spartina Alterniflora	86.31	80.90	**86.85**	83.16	74.94	76.07	82.72	81.73	83.26	83.11
OA	93.89	86.19	83.54	85.40	86.67	87.37	88.10	88.37	89.81	**90.69**
AA	92.15	77.90	77.03	76.58	79.77	80.55	80.01	81.16	82.80	**85.09**
Kappa	0.9231	0.8279	0.7957	0.8184	0.8335	0.8422	0.8512	0.8547	0.8724	**0.8834**

be more biased toward samples with more categories. The classifier's attention is focused on the categories with large samples, so that the categories with a small number are not paid attention to, which will lead to poor classification effects for samples with a small number of categories. It can be seen from the experimental results that the classification accuracy of a large number of categories, such as Healthy Grass and Synthetic Grass, is generally higher than that of Residential, Road, and Railway. Under the condition of sample imbalance, the algorithm based on transfer learning is better than the baseline method for the correct recognition of small samples. The algorithm proposed in this article, C-ours, has performed probability adaptation on the basis of feature alignment, so it has reached the best effect in most categories. The method can generate more distinguishing features than the baseline methods, and the effect is even better than the B-HSI classification results. It can be seen from Fig. 5 that in the shaded area, C-ours also has a better performance, and the classification results are more reasonable. This shows that the C-ours method that adds a mapping structure during probability adaptation to maintain can learn more discriminative feature representations and find better decision boundaries.

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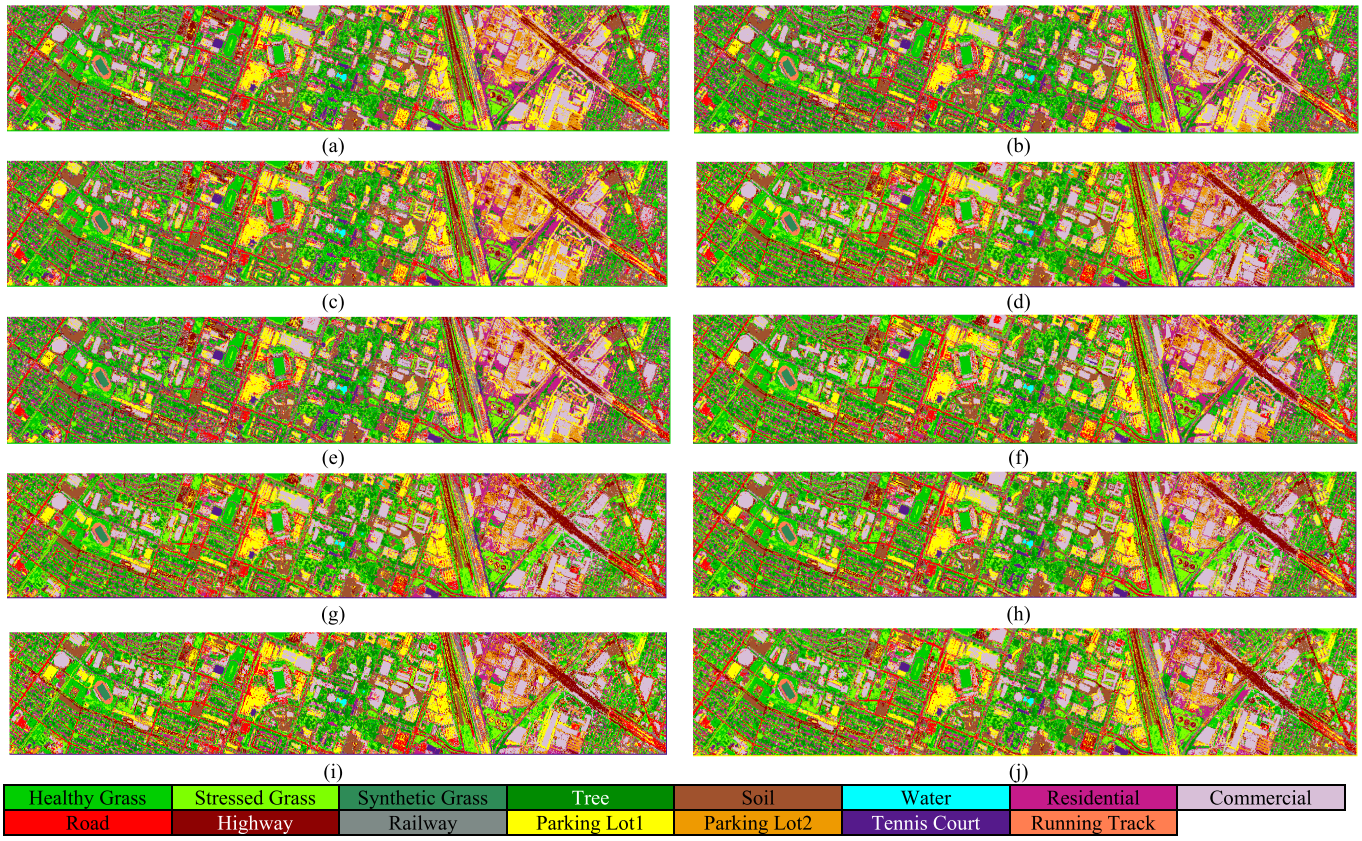


Fig. 5. Classification results on UH dataset. (a) B-HSI. (b) B-MSI. (c) JSLoL. (d) CGCDL. (e) CDLS. (f) Cospace. (g) J-CDLS. (h) C-CDLS. (i) J-ours. (j) C-ours method.

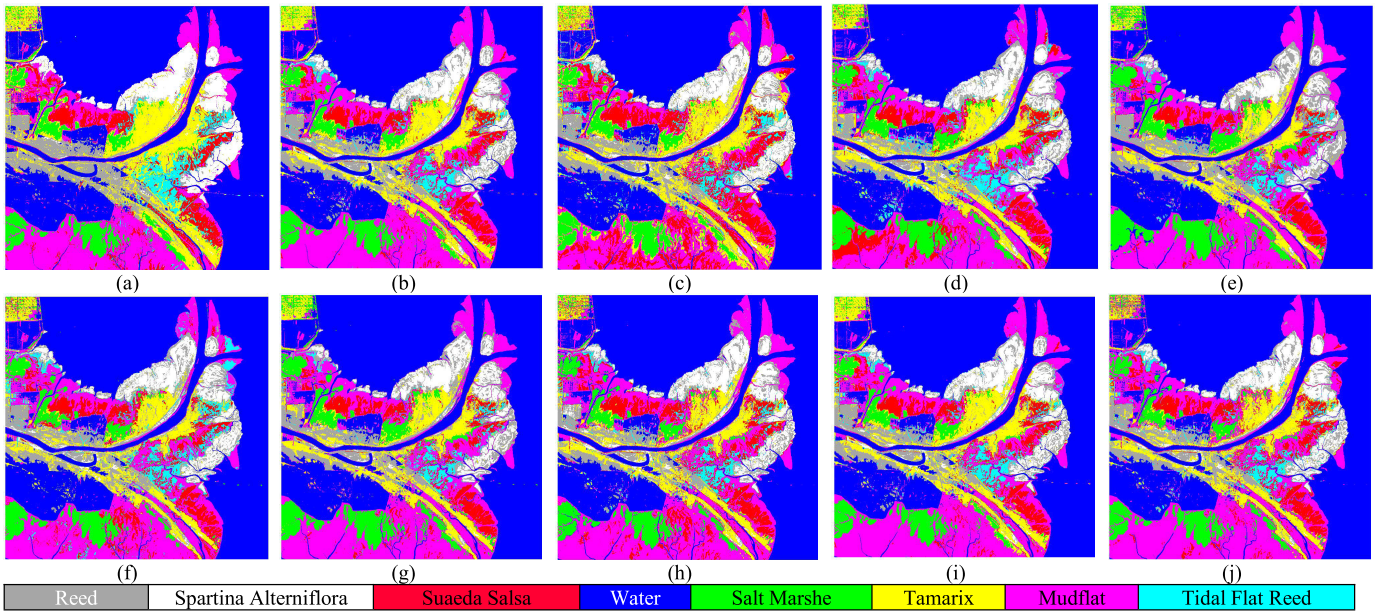


Fig. 6. Classification results on Yellow River Estuary_1 dataset. (a) B-HSI. (b) B-MSI. (c) JSLoL. (d) CGCDL. (e) CDLS. (f) Cospace. (g) J-CDLS. (h) C-CDLS. (i) J-ours. (j) C-ours method.

Similarly, in order to make the comparison easier visually, this article also gives the results of different methods using the NN classifier on the two Yellow River Estuary datasets, as shown in Figs. 6 and 7. It can be seen from Figs. 6 and 7 that the classification result is relatively poor due to the lack of spectral information of the MS image. The reconstruction-based MS–HS classification methods, JSLoL and CGCDL, introduce HS information into MS, but there are differences

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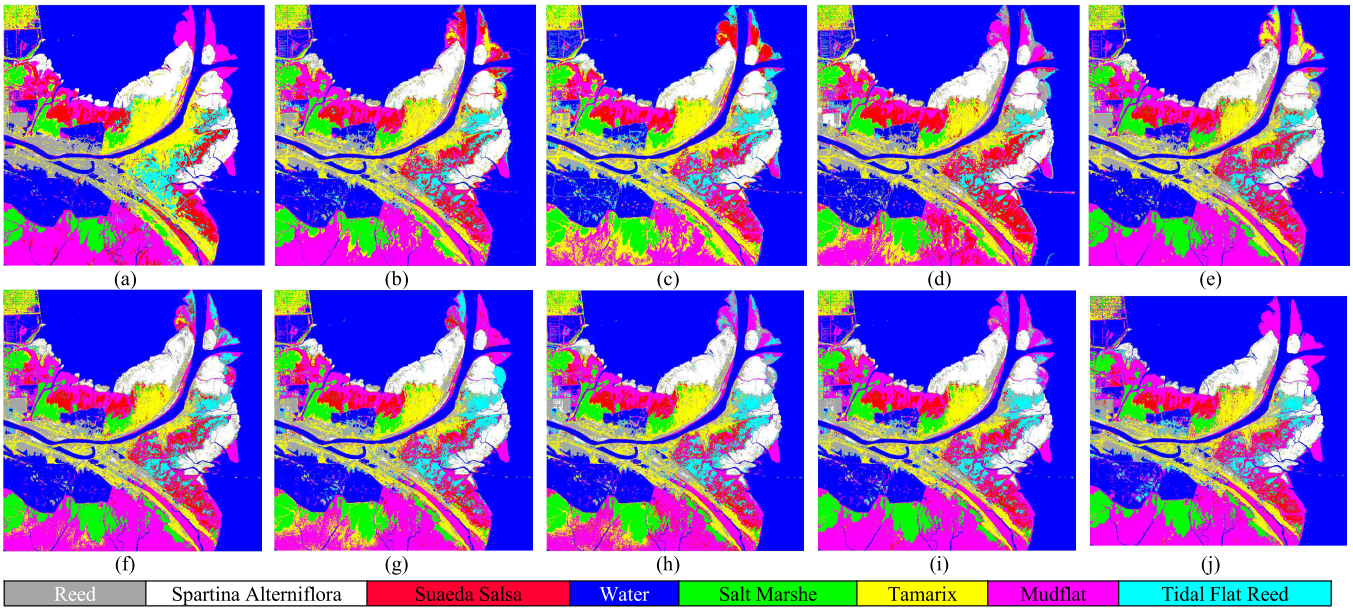


Fig. 7. Classification results on Yellow River Estuary_2 dataset. (a) B-HSI. (b) B-MSI. (c) JSLoL. (d) CGCDL. (e) CDLS. (f) Cospace. (g) J-CDLS. (h) C-CDLS. (i) J-ours. (j) C-ours method.

with real HS data, and the effect is lower than the results of B-MSI classification. On the two Yellow River Estuary datasets, the MS–HS classification method based on transfer learning is superior to the results of B-MSI classification by adapting the probability of spectral features. Compared with these collaborative classification methods, J-ours and C-ours not only align the feature space of MS and HS data but also align the feature distribution of the two. C-ours adds two constraints of discriminant sparse coefficient and classification error when generating HS images, so it achieves the best results. The learned features can more effectively identify categories with similar features in MS data, such as Spartina Alterniflora, Reed, and Tamarix. As shown in Figs. 6 and 7, both the structural information and the detailed texture are more similar to the classification map of the real HS images. Our method can make the classification map relatively smooth in some categories, such as Mudflat and Reed in the datasets. Therefore, the classification result of the proposed method is more reasonable.

As shown in Figs. 8–10, in order to illustrate the effectiveness of the proposed method, the output of different methods is visualized. Figs. 8–10 show the t-distributed stochastic neighbor embedding (t-SNE) visualization of features of the three datasets using different methods. B-HSI and B-MSI only involve one domain, where "." represents the spectral features of HS images and "+" represents those of MS images. What is more, other methods involve two domains, where "." represents the source domain and "+" represents the target domain. It can be seen from Figs. 8–10 that the Cospace method does not adapt features between HS and MS images. In the Cospace method, the visualization features of HS and MS images maintain their original shapes. However, CDLS has probabilistically adapted spectral features of HS images and MS images, but the effect is not very good, such as the Synthetic Grass and Trees categories in the Houston dataset. There is still a big difference in the distribution between the real HS image (source domain) and the generated HS image (target domain) in the JSLoL and CGCDL methods. The J-CDLS and C-CDLS methods probabilistically adapt the generated HS image and the real HS image, but the distance between the same category is still very large, such as the Synthetic Grass and Healthy Grass categories in the Houston dataset. The J-ours and C-ours methods increase the distance between classes and effectively reduce the distance within classes, such as the Salt Marshe and Tamarix categories in the Yellow River Estuary_1 and the Yellow River Estuary_2 datasets. Therefore, most samples in each class of the C-ours method are clustered with a smaller intraclass distance, which will better adapt the source and target domains; thus, the proposed method obtains better performance.

In conclusion, the performance of the proposed method is far superior to other methods. More specifically, on the Houston dataset, the classification accuracy obtained by J-ours and C-ours is even higher than the baseline B-HSI. Although compared with other methods, J-ours and C-ours do not always perform best in every category. However, on the two Yellow River Estuary datasets, the classification accuracy obtained by the C-ours method ranks first in most categories and second in other categories. The Houston dataset has similar conclusions, and these all prove the effectiveness of the proposed method.

D. Parameter Sensitivity Analysis

In the proposed method, the quality of image reconstruction is closely related to the result of probabilistic adaptation and the performance of classification. The relevant parameters of image reconstruction have been analyzed in JSLoL and CGCDL methods, and these methods

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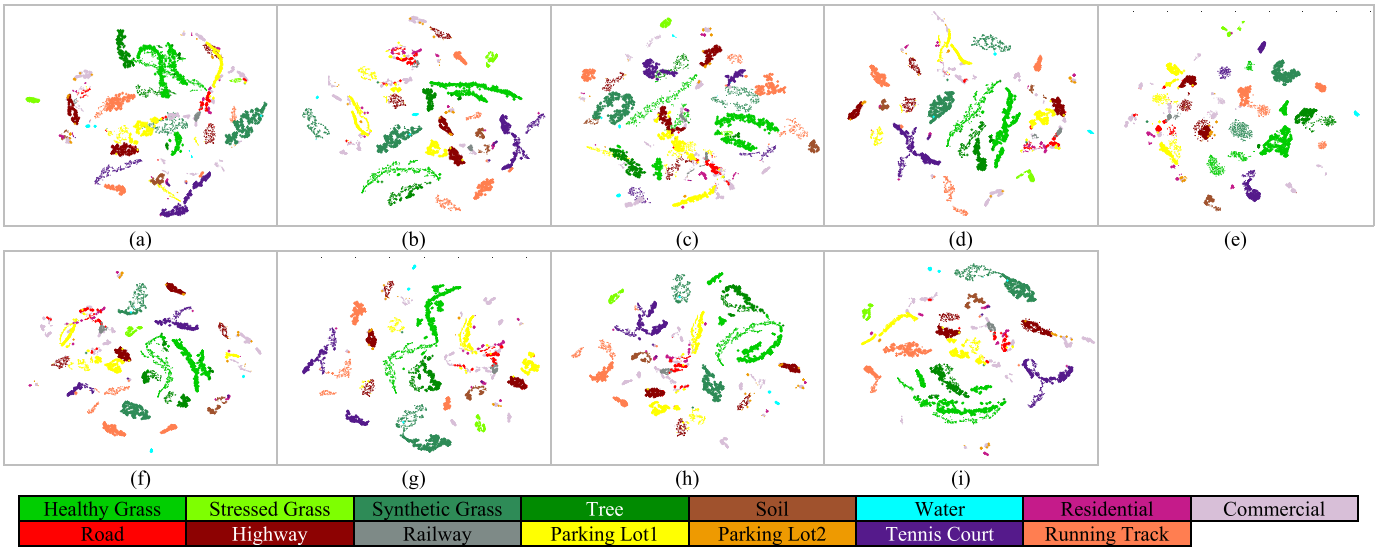


Fig. 8. T-SNE visualization on the Houston dataset. (a) B-HSI and B-MSI. (b) JSLoL. (c) CGCDL. (d) CDLS. (e) Cospace. (f) J-CDLS. (g) C-CDLS. (h) J-ours. (i) C-ours method.

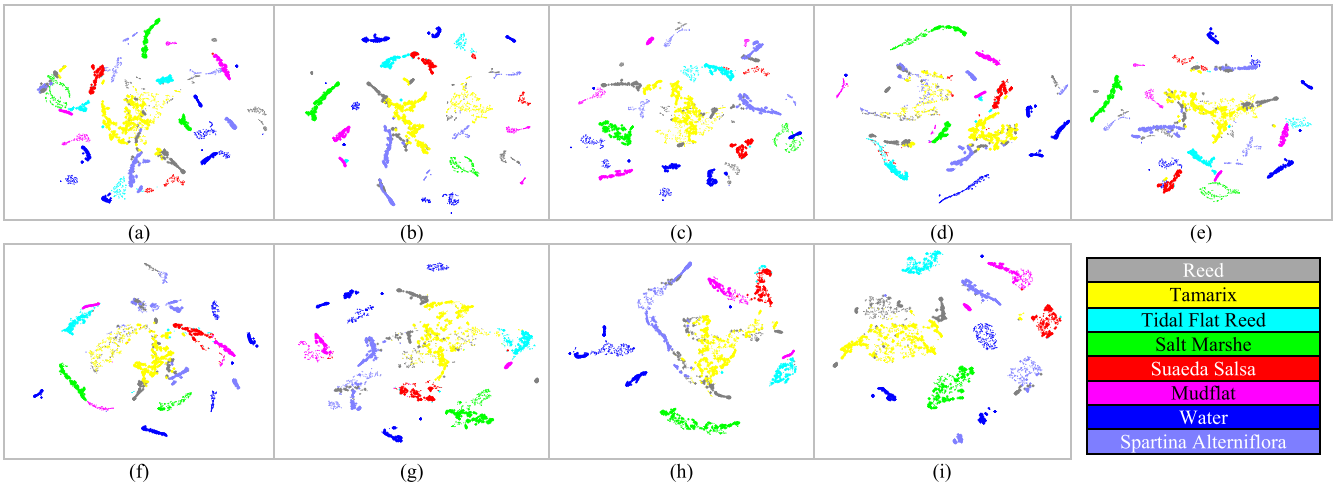


Fig. 9. T-SNE visualization on the Yellow River Estuary_1 dataset. (a) B-HSI and B-MSI. (b) JSLoL. (c) CGCDL. (d) CDLS. (e) Cospace. (f) J-CDLS. (g) C-CDLS. (h) J-ours. (i) C-ours method.

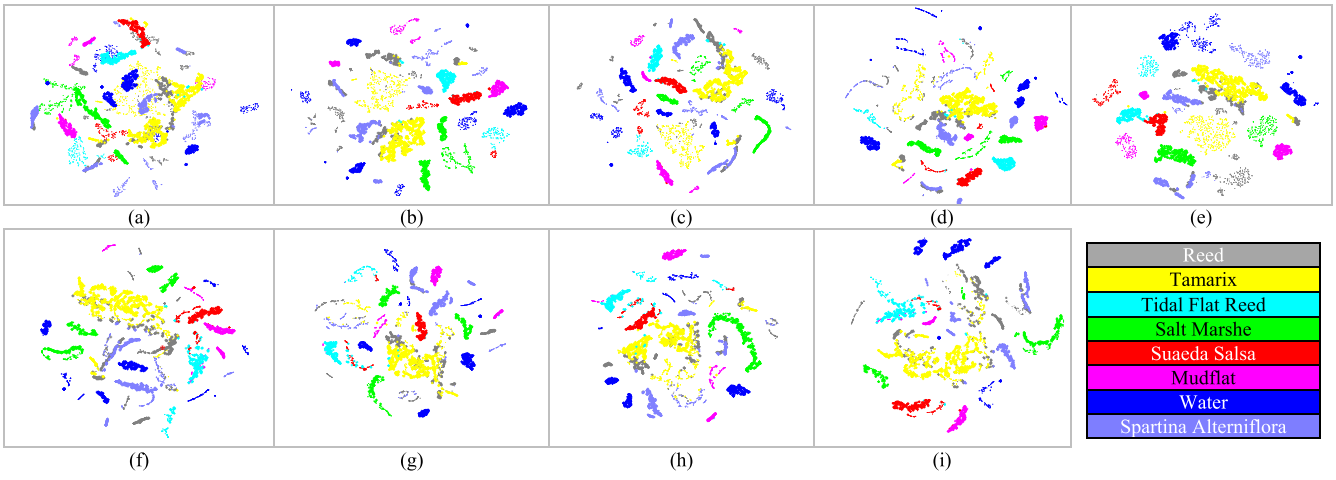


Fig. 10. T-SNE visualization on the Yellow River Estuary_2 dataset. (a) B-HSI and B-MSI. (b) JSLoL. (c) CGCDL. (d) CDLS. (e) Cospace. (f) J-CDLS. (g) C-CDLS. (h) J-ours. (i) C-ours method.

also find that the parameter settings in this article are relatively stable, so this article applies their parameter settings to our datasets. In order to further analyze the sensitivity of the regularization parameter λ, other parameters during

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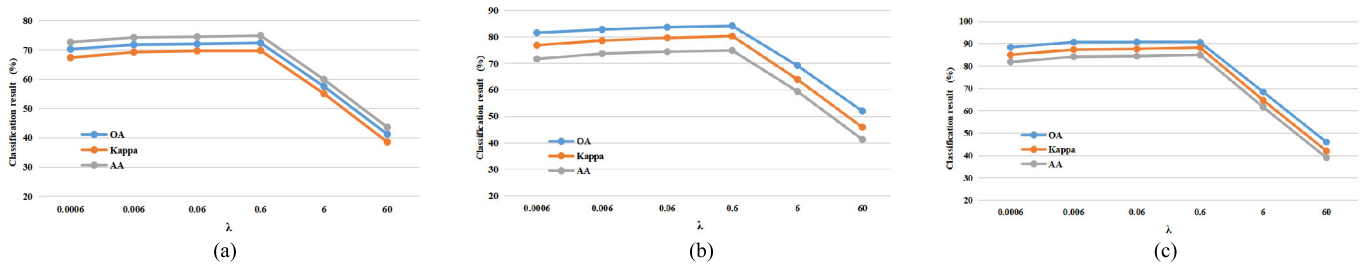


Fig. 11. Parameter sensitivity analysis on three datasets. (a) Houston dataset. (b) Yellow River Estuary_1 dataset. (c) Yellow River Estuary_2 dataset.

image reconstruction are fixed and λ is varied in the range of $\{6 \times 10^{-4}, 6 \times 10^{-3}, \ldots, 6 \times 10^{0}, 6 \times 10^{1}\}$. We use the C-ours method to adjust the regularization parameter λ for parameter-sensitive analysis experiments and use the OA, AA, and kappa coefficient (κ) to evaluate the classification performance. In Fig. 11, the relationship between the classification performance and the regularization parameter is shown. From Fig. 11, it can be found that OA, AA, and kappa have the same trend in the three datasets as the parameters increase. At the beginning, as the parameter λ value increases, the classification performance gradually improves. When the parameter is equal to 0.6, the classification effect is the best, and then, the classification performance drops sharply. From Fig. 11, it can be seen that fixing $\lambda = 0.6$ would also be a reasonable choice. Therefore, this article fixes $\lambda = 0.6$ in the experiments.

IV. Conclusion

It is meaningful to study how to use limited HS data to improve the classification ability of large area MS data. In response to this problem, this article proposes an MS–HS image collaborative classification method, which reduces the difference between the generated HS data and the real HS data. Specifically, while performing image reconstruction by coupled dictionary learning, this method also performs conditional probability adaptation of spectral features and adds constraint items about spectral mapping, thus improving the classification ability of MS images. In this article, experiments have been conducted on three cross-modal MS–HS datasets using different algorithms, and the experimental results of J-ours and C-ours prove its superiority and validity to MS-HS data. In future work, we will solve the difference in spatial resolution while solving the spectral difference and realizing the use of MS–HS images with different spatial–spectral resolutions to improve the classification ability of MS images.

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